Calcium Gluconate

FORM

Ampoule of 10% calcium gluconate injection solution contains 1g of calcium gluconate = 2.2mmol of Ca^{2+} in 10ml.

INDICATION

1. Treatment of neonatal tetanus or arrhythmias
2. Treatment of hyperkalaemia
3. Treatment of acute hypocalcaemia

DOSE RANGE

1. **Treatment of neonatal tetanus or arrhythmias**
2. **Treatment of hyperkalaemia - USE WITH ECG MONITORING **DISCUSS WITH CONSULTANT BEFORE USE****

AGE	DOSE	FREQUENCY	ROUTE
Birth to 6 months	0.11mmol/kg	SINGLE DOSE May be repeated	**Slow Injection over 5-10 min.**

3. **Treatment of acute hypocalcaemia (Calcium <1.8mmol/L or ionised Calcium <0.7mmol/l) - USE WITH ECG MONITORING **DISCUSS WITH CONSULTANT BEFORE USE****

AGE	DOSE	FREQUENCY	ROUTE
Birth to 6 months	0.11- 0.22mmol/kg*	SINGLE DOSE	**Slow I.V. INFUSION see below**

* In cases of severe symptoms of hypocalcaemia in neonates or infants, higher initial doses of up to 0.44mmol/kg may be necessary for a quick restoration of a normal serum calcium level.

RECONSTITUTION

Already in solution

DILUTION

Calcium Gluconate 10% inj.	10ml
Sodium chloride 0.9% inj.	Up to 50ml total

Gives Calcium Gluconate 0.044mmol in 1ml (2%) solution. Use the required volume.

Dosage guide (using diluted solution)

0.11mmol/kg equivalent to 2.5ml/kg
0.22mmol/kg equivalent to 5ml/kg

METHOD OF ADMINISTRATION

IV infusion at a rate not exceeding 30ml/hr, of the diluted solution above.

- Note: In emergencies doses of 0.11mmol/kg may be given undiluted as calcium gluconate 10% solution via a central venous access device but due to the high osmolality of undiluted calcium gluconate it may cause venous irritation and tissue damage in cases of extravasation.

Neonatal

COMPATIBILITY

Solution compatibility	Glucose 5%, Glucose 10%, Sodium Chloride 0.9%
Solution incompatibility	All other solutions
IV Line compatibility	Aciclovir, Adrenaline, amiodarone, (in glucose only), caffeine citrate, cefotaxime, dexmedetomidine, fentanyl, furosemide, ganciclovir, gentamicin, heparin, insulin, isoprenaline, metronidazole, midazolam, milrinone, morphine, noradrenaline, potassium chloride, sodium chloride 0.9%, vancomycin (glucose only), vercuronium
IV Line incompatibility	Calcium salts can form complexes with many drugs, and this may result in a precipitate. Amphotericin B, bicarbonates, carbonates, ceftriaxone, citrates, dobutamine, phosphates, sulfates, tartrates and tetracyclines.

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS

- See Summary of Product Characteristics and most recent edition of BNF for Children (links below)

FURTHER INFORMATION

- Extravasation risk – extravasation may cause tissue irritation and necrosis. Soft tissue calcification due to extravasation of calcium solutions
- Calcium administration may be associated with bradycardias and other cardiac arrhythmias and patients should be monitored accordingly.
- Please note for the brand **Hameln Pharma Ltd** a maximum dose of 0.22mmol/kg is recommended for children under 18 year old due to the risk of exposure to aluminium.
- Care should also be taken to correct any associated hypomagnesaemia
- Concomitant use of ceftriaxone and intravenous calcium-containing products is contraindicated in neonates (<28 days of age). Ceftriaxone should not be used in neonates (<28 days of age) if they are receiving (or are expected to receive) calcium-containing intravenous products.

STORAGE

- Calcium Gluconate Injection may be stored at room temperature and should be protected from light.If a precipitate appears, discard solution.

PH

5.5 – 7.5.

LICENSED STATUS

Licensed medicine. Doses used in monograph off label use.

APPLICABLE POLICIES

West of Scotland Neonatal Guidelines

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department